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Sub-millimeter molybdenum and uranium isotopes track millennial redox events in the Cambrian ocean



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Submillimeter-scale redox archives in organic-rich mudrocks elucidate dynamic environmental stress shaping early marine animal ecosystems. During the late Cambrian, benthic animals, including trilobites, thrived in intermittently oxygen-deficient (predominantly euxinic) outer shelf environment where the astronomically-tuned Scandinavian Alum Shale was deposited. We present ultrahigh-resolution (0.25 mm) molybdenum and uranium isotope ($\delta^{98}\text{Mo}$ - $\delta^{238}\text{U}$) and concentration profiles, revealing millennial-scale redox fluctuations that intermittently disrupted bottom-water euxinia. Varying ocean circulation, redox buffer supplies, and/or reduced productivity likely created short-lived aerobic niches for opportunistic benthic animals. Distinct $\delta^{98}\text{Mo}$ - $\delta^{238}\text{U}$ patterns indicate recurrent chemocline shoaling with toxic hydrogen sulfide incursions into surface waters, resembling metal drawdown in modern restricted basins. These fluctuations punctuated a million-year-long global oceanic anoxic event linked to the Steptoean Positive Carbon Isotope Excursion. Following this excursion, local redox conditions oscillated around a globally more oxygenated baseline, coinciding with animal diversifications in Baltica and worldwide. These millennial-scale redox changes underscore the need for higher-resolution geochemical approaches in paleoenvironmental reconstructions.

Episodes of organic-rich shale deposition occurred throughout Earth's history^{1–6} with a notable abundance in the early Paleozoic when anoxic marine zones were more widespread and atmospheric pO_2 likely was lower than today^{7,8}. Organic-rich shales are primarily deposited in oxygen-poor environments, either under euxinic conditions where hydrogen sulfide (H_2S) is present in the anoxic bottom waters, under ferruginous conditions (rare in the modern ocean) where anoxic bottom waters are sulfur-limited, or under low-oxygen conditions, where the bottom water O_2 levels are significantly lower than in the surface ocean^{9–14}. Redox fluctuations at the seafloor are expected to exert first-order control on habitability and perhaps also on biotic evolution, since animals are obligate aerobic organisms and H_2S is toxic to essentially all eukaryotes. The redox conditions near the seafloor fluctuate because marine oxygen supply and consumption vary over a wide range of timescales from seasonal to millennial to millions of

years^{12,15–17}. Yet, rapidly changing redox conditions in the oceans challenge the interpretation of sedimentary paleoredox proxies extracted from bulk rock samples that may aggregate time over thousands to millions of years, highlighting the need for obtaining records at finer stratigraphic resolution with an accurately tuned time scale.

The late Cambrian Alum Shale in Scandinavia was deposited under overall euxinic conditions and, perhaps occasionally, ferruginous conditions^{10,18–20}. However, like many other anoxic shales in the world^{21,22}, the Alum Shale also bears evidence of benthic animal activity, including abundant trilobite communities^{23–26} (Fig. 1), indicative of periods with aerobic conditions at the seafloor. The Alum Shale offers opportunities to investigate in detail how redox conditions fluctuated at millennial time scales in the Cambrian ocean. Episodic oxygenation events have been suggested to have opened niches for bottom-dwelling animals¹⁵.

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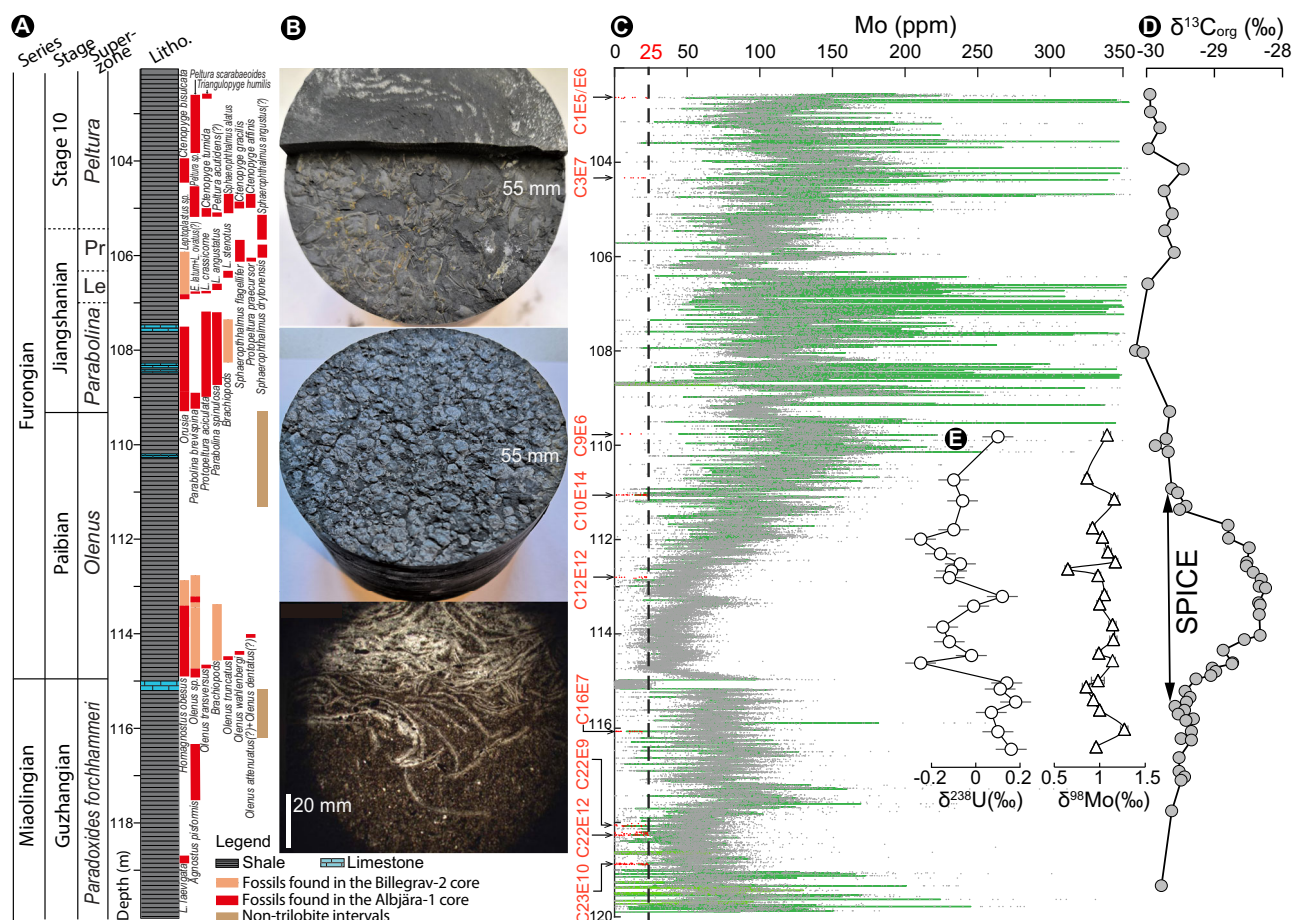


Fig. 1 | Location, stratigraphy, and geochemical profile of the Billegrav-2 core (Bornholm, Denmark). **A** Stratigraphy, lithology, and fossils. The detailed biozone framework is established based on existing fossil records and further constrained by gamma log correlation with other fossiliferous Alum Shale cores^{36,76,77}. Biozone abbreviations: Pr *Protopenetula*, Le *Leptoplastus*. **B** Photos of fossils found in the Alum Shale Formation, e.g., *Olenus transversus* trilobites (top) from the lower part of the *Olenus* Superzone, brachiopods (middle), and cross-section of *Orusia*

lenticularis (bottom) from the *Parabolina* Superzone^{15,36}. **C** High-resolution (0.2 mm) Mo concentration profile with stratigraphic levels of ten MREs investigated in this study. The gray dots and green curve represent XRF Mo concentrations calibrated against ICP-MS data¹⁵ and a LOESS fitted curve, respectively. **D** Organic carbon isotope profile across the SPIKE event in the Billegrav-2 core⁷⁶. **E** Million-year-scale $\delta^{98}\text{Mo}$ and $\delta^{238}\text{U}$ data from hand-size Alum Shale specimens across the SPIKE event¹⁰.

Independent evidence for millennial oxygenation events in the Alum Shale Sea was inferred from recurrent decreases in sedimentary molybdenum (Mo) concentrations from well above 100 ppm to below 25 ppm detected at sub-millimeter stratigraphic resolution¹⁵ (0.2 mm; Fig. 1C). In addition, we here cut samples at sub-millimeter resolution and use isotopic proxies to advance previous efforts and improve reconstructions of the processes governing redox dynamics at the millennial scale.

Molybdenum and uranium concentrations are powerful paleoredox proxies, and so are their stable isotope compositions, $\delta^{98}\text{Mo}$ and $\delta^{238}\text{U}$. Both elements occur as dissolved anionic compounds with a conservative behavior in oxic seawater, which is reflected in their long residence times in the Cambrian (~100 kyr)^{10,27,28} and modern ocean (~400 kyr)^{29,30}. Both elements are preferentially enriched in sediments deposited under reducing bottom water conditions³¹. However, the link between Mo and U concentrations and the local redox conditions that control their removal is not straightforward, since muted enrichments can also occur in basins with limited supply of these metals. For example, the seawater supply of Mo and U from the open ocean might be outpaced by rapid removal into anoxic and sulfidic sediments within a basin. Additionally, the inventories of Mo and U in the open ocean would have been lower at times when marine anoxia was globally more widespread^{10,20}. Paired $\delta^{98}\text{Mo}$ and $\delta^{238}\text{U}$ analyses in sediments have opened prospects of discriminating between these scenarios^{10,32–34}; however, to our knowledge, they have not yet been investigated at the

submillimeter scale. Here, we re-examined ten “Millennial Redox Events” (MRE, consisting of both oxygenation or deoxygenation scenarios) previously described as “Brief Oxygenation Events” from the Billegrav-2 Alum Shale core obtained on Bornholm, Denmark¹⁵ (see location in Supplementary Fig. 1), by applying paired U and Mo concentrations and stable isotope analyses at the submillimeter scale (as low as 0.25 mm stratigraphic resolution). The Billegrav-2 core has been chronologically dated using cyclostratigraphy³⁵ anchored in the geological record³⁶, providing interpolated estimates for the average durations of the MREs spanning many centuries to several millennia (~650–8000 yrs). These durations are interpreted by interpolating 405 kyr-eccentricity cycles over approximately 80 cm of stratigraphy, but the events may be significantly shorter, particularly if extended periods of non-deposition were dominant.

Molybdenum and uranium isotopes

The long residence times of U and Mo in the Cambrian ocean are much longer than the ocean mixing time of ~1.5 kyr³⁷, implying that oceans are well mixed with respect to Mo and U and that significant changes in the global ocean inventories of these elements occur on timescales of over ~10⁵ yrs. Therefore, millennial-scale changes in Mo and U concentrations and their isotope compositions (here, expressed over millimeters of stratigraphy) in sedimentary rocks must reflect changes in the local environment rather than changes in global seawater composition or metal inventories. In

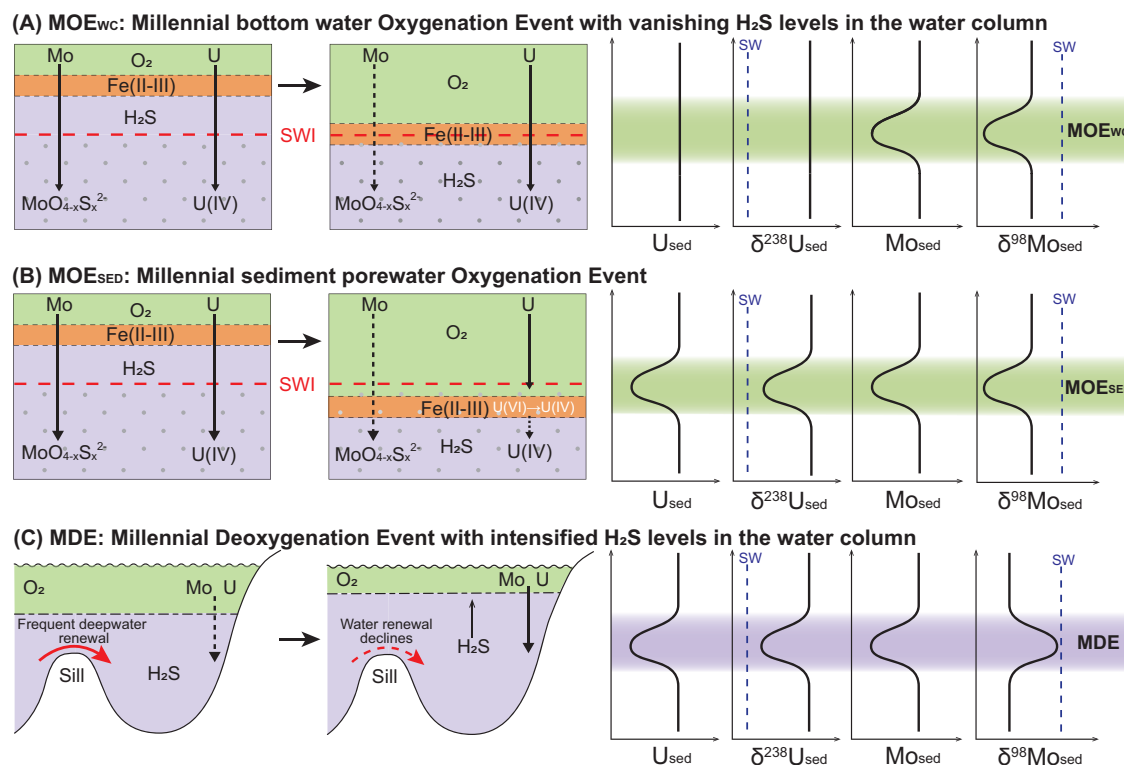


Fig. 2 | Conceptual model illustrating the expected changes of the sedimentary U, Mo, $\delta^{98}\text{Mo}$, and $\delta^{238}\text{U}$ in three MRE scenarios. The zones in green, orange, and purple represent oxic, anoxic (where Fe(III), Mn(IV), and nitrate reduction dominates), and sulfidic (where sulfate reduction prevails) zones, respectively. Black solid and dashed arrow lines represent relative more or less Mo and U fluxes to the sediment, respectively. SWI sediment-water interface, SW sea water. **A** During a Millennial Oxygenation Event in the water column (MOE_{WC}), Mo burial on settling particles will be dramatically reduced and lower $\delta^{98}\text{Mo}$ values in the sediments (greater Mo isotope offset from contemporaneous seawater) are expected, whereas most U burial likely occurs below sediment-water interface where conditions are

permanently sulfidic, resulting in smaller changes. **B** During a more extensive oxygenation event that also introduces O₂ into the sediments (MOE_{SED}), U and $\delta^{238}\text{U}$ will also be affected. The U isotope offset in oxic settings is smaller than under euxinic conditions, which leads to lower $\delta^{238}\text{U}$ closer to crustal composition and oceanic input. **C** During a Millennial Deoxygenation Event (MDE) where H₂S levels in the water column rises and cause removal of Mo and U via settling particles to the sediments, which will draw down the Mo and U inventory in the overlying water column (given slow replenishment) resulting in low sedimentary Mo and U concentrations and sedimentary $\delta^{98}\text{Mo}$ and $\delta^{238}\text{U}$ compositions approaching seawater values.

euxinic basins, Mo and U are efficiently scavenged and buried^{18,38,39}, whereas burial rates are significantly lower in suboxic environments, where sulfide accumulates at depth inside the sediments. Figure 2 outlines three conceptually distinct MRE scenarios and how the sedimentary $\delta^{98}\text{Mo}$, $\delta^{238}\text{U}$, Mo, and U concentrations would be expected to change when the supply or burial rates vary in the basin.

As a first scenario, a redox-stratified euxinic basin interrupted by periodic millennial-scale bottom water oxygenation or vanishing H₂S levels in the water column (MOE_{WC}; Fig. 2A) will slow Mo sequestration from the water column into the sediments. In anoxic and euxinic settings, dissolved Mo (MoO_4^{2-}) reacts with H₂S to form thiomolybdates ($\text{MoO}_{4-x}\text{S}_x^{2-}$, $x=1-4$), which are removed through precipitation with sinking iron sulfides and organic matter^{38,40,41}. Each step of the conversion from molybdate towards tetrathiomolybdates is associated with stable isotope fractionation^{18,42,43}. When H₂S level exceeds 11 μM , Mo will be quantitatively removed via particle scavenging into the sediments⁴⁴, producing sedimentary $\delta^{98}\text{Mo}$ compositions approaching that of the overlying seawater⁴⁵⁻⁴⁷. However, in weakly euxinic or intermittently oxygenated water columns (H₂S < 11 μM or fluctuating), Mo removal can be incomplete, yielding a negative $\delta^{98}\text{Mo}$ offset in sediments relative to seawater, because the particle reactive Mo fraction in the basin consists of isotopically fractionated thiomolybdate⁴⁸. Unlike Mo, uranium sequestration begins at around the Fe(II)–Fe(III) redox boundary in the absence of H₂S^{33,49,50} and continues through the sulfidic zone. Additionally, reductive U sequestration typically occurs in the uppermost few centimeters of the sediments where soluble U(VI) from seawater diffuses across the sediment-water interface⁵¹,

whereas Mo removal mainly occurs at the O₂/H₂S chemocline. Therefore, the MOE_{WC} scenario could leave the U concentration and $\delta^{238}\text{U}$ offset relative to overlying seawater largely unaffected as long as the pore fluids remained anoxic and sulfidic at depth inside the sediments. Given the differences, the coupling using Mo and U elemental abundances and isotopes has the potential to more precisely bracket the oceanic redox conditions.

In a second and more extensive oxygenation scenario, where the sediment porewaters also become oxygenated to a significant depth (MOE_{SED}; Fig. 2B), we expect sedimentary U and $\delta^{238}\text{U}$ to change. With oxygenated porewaters, lower U-enrichments and lower $\delta^{238}\text{U}$ are expected, as seen in modern sediments deposited under oxic waters with sulfide accumulating at depth^{51,52}. In this scenario, the $\delta^{238}\text{U}$ composition of the sediments will typically be closer to coeval seawater with a smaller isotopic offset than observed in euxinic settings ($\sim 0.6\text{--}0.8\text{‰}$ ^{33,53}). If Mo and U were removed via manganese oxides during MOE_{SED} events, this would also be accompanied by lower $\delta^{98}\text{Mo}$ values and lower $\delta^{238}\text{U}$ values^{54,55}.

As a third and quite opposite scenario, we propose a Millennial Deoxygenation Event (MDE; Fig. 2C), where Mo and U are removed even faster and more completely in the basin relative to the basal supply from the open ocean (e.g., declining deepwater renewal rate). This rapid removal would eventually lower the sedimentary Mo and U concentrations. As stated above, the oceanic Mo and U inventories cannot decrease significantly over such short time scales. During millennial-scale intensification of euxinia, elevated H₂S concentrations and/or shoaling of the O₂/H₂S chemocline in the water column would enhance the

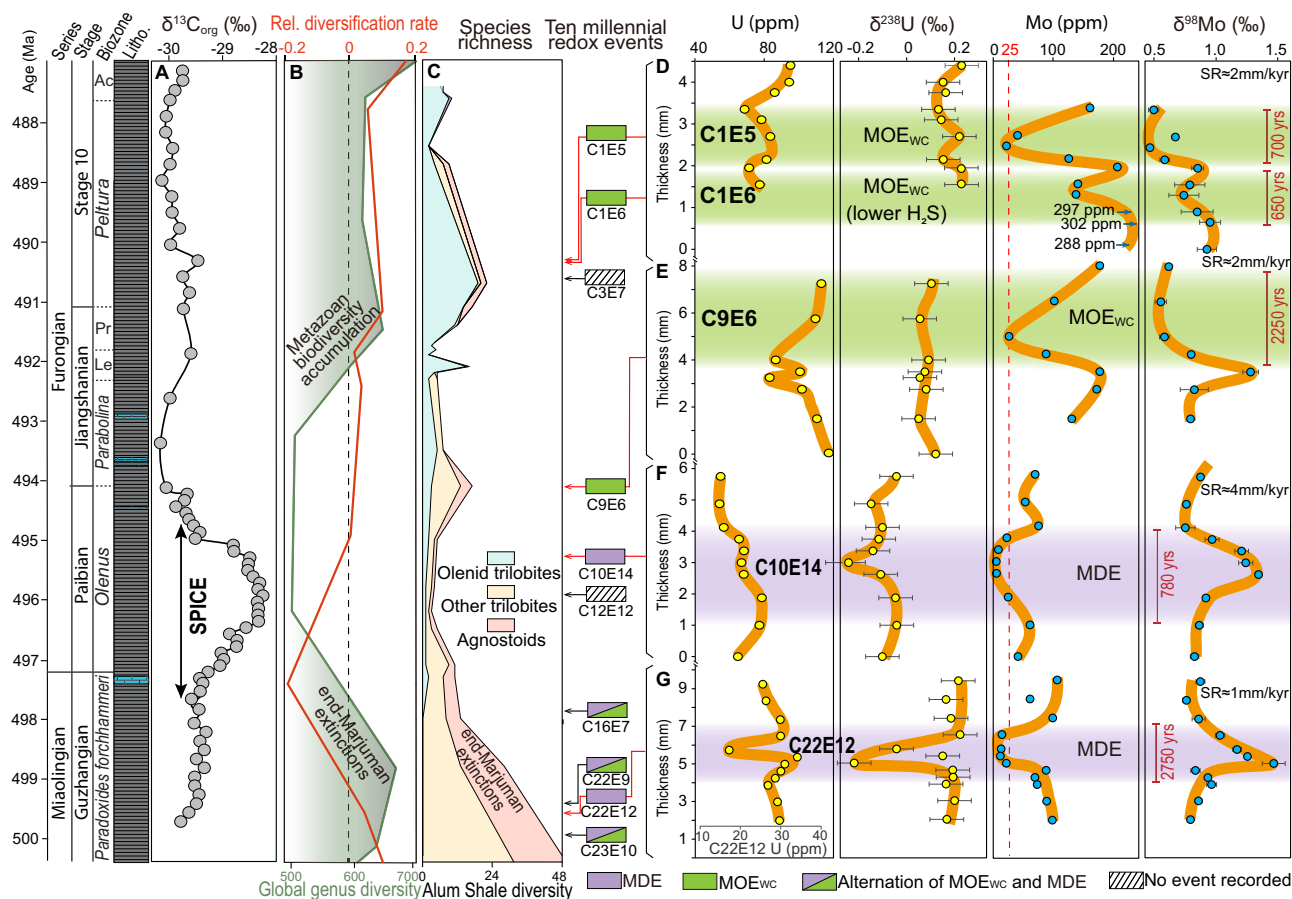


Fig. 3 | Stratigraphic levels and geochemical characteristics of studied events associated with richness trajectories of marine fossils. **A** Stratigraphy, lithology, and carbon isotopes of the Alum Shale Formation in the Billegrav-2 core. **B** Global genus-level diversity and relative diversification rate^{74,75}. **C** Species diversity in the Alum Shale Formation, see data in Supplementary Table 2. **D–G** Abundances and

isotope compositions of U and Mo at sub-millimeter resolution in the Alum Shale for five representative MREs. The geochemical profiles of the remaining five events are provided in Supplementary Fig. 2. Interpolated sedimentation rates (SR) are from Sørensen et al.³⁵.

formation of thiomolybdates, facilitating Mo removal via particle settling into the sediments^{38,40}. Rising levels of aqueous H_2S would, in general, imply that H_2S also migrated towards the ocean surface, i.e., chemocline shoaling. At high enough $[\text{H}_2\text{S}]$ ($>11\ \mu\text{M}$), quantitative Mo removal produces sediments with higher $\delta^{98}\text{Mo}$ compositions close to that of the seawater source (i.e., open ocean) in a fashion similar to the trend observed with basinal restriction within euxinic basins today^{45,47}. In the MDE scenario, we also expect the U isotopic offset between the water column and the sediment to diminish and the sedimentary $\delta^{238}\text{U}$ to approach the composition of seawater input to the basin (Fig. 2C). Thus, the MDE is expected to produce a negative correlation of $\delta^{238}\text{U}$ and $\delta^{98}\text{Mo}$, as observed in modern basins with sluggish ventilation³³.

Results and discussion

A total of 127 Alum Shale samples from ten MREs of the Billegrav-2 core display U, $\delta^{238}\text{U}$, Mo, and $\delta^{98}\text{Mo}$ values in ranges of 17–179 ppm, -0.27 – 0.30‰ , 6–302 ppm, and 0.47 – 1.83‰ , respectively (Supplementary Table 1). Five representative MREs are shown in Fig. 3 and described in the following text, and the remaining five MREs are discussed in the Supplementary Note 3. All ten MREs are classified as either MOE_{WC} or MDE; no MOE_{SED} events were identified. For C10E14, we cut an additional sample with a direction perpendicular to the bedding plane, which could represent a bulk rock sample of this event. This sample contains Mo and U concentrations and $\delta^{98}\text{Mo}$ and $\delta^{238}\text{U}$ compositions of 59 ppm, 61 ppm, $0.78 \pm 0.06\text{‰}$ and $-0.05 \pm 0.07\text{‰}$, respectively. The sub-millimeter bedding-parallel samples demonstrate significant variation of these proxies

across the same stratigraphic range, with ranges of 6–76 ppm, 53–78 ppm, 0.75 – 1.35‰ , and -0.24 to -0.04‰ ($n = 10$). This comparison demonstrates the substantial risk of missing short-term signals when using bulk rock analyses.

Bottom water oxygenation at millennial scale (MOE_{WC}) and its drivers

The data patterns observed in C1E5, C1E6, and C9E6 are consistent with the expected characteristics of the MOE_{WC} scenario (Figs. 2A and 3DE). These events contain $\delta^{98}\text{Mo}$ decreases across stratigraphic thicknesses of 1.3–4.5 mm (i.e., lasting 650–2250 yrs; calculated based on ref. 35, which all correspond with a drop in Mo concentrations (from >170 to <25 ppm). Although we find evidence of increasing Al concentration that serves as a proxy for clastic inputs, the decreasing Mo concentrations cannot be explained only by sediment dilution due to changes in sedimentation rate, since the change of Al concentration is modest (Supplementary Fig. 3). Also, the Alum Shale mud was deposited exceedingly slowly (~ 1 – $4\ \text{mm}\ \text{kyr}^{-1}$ post-compaction^{35,36}), and even euxinic sediments deposited $\sim 10^4$ times faster also capture high Mo concentrations^{14,56}. Additionally, U and Mo concentrations are exceptionally higher than those of upper continental crust (2.7 ppm U and 1.1 ppm Mo⁵⁷), suggesting that the signals are authigenic rather than detrital in origin. Unlike the Mo-based proxies, the sedimentary $\delta^{238}\text{U}$ values vary by only ~ 0.1 – 0.2‰ within a single MOE_{WC} event (Fig. 3). Pyrite is present throughout the studied intervals¹⁵, thus we can infer that the sedimentary porewaters were persistently anoxic and sulfidic at depth with reductive U removal as the dominant U removal

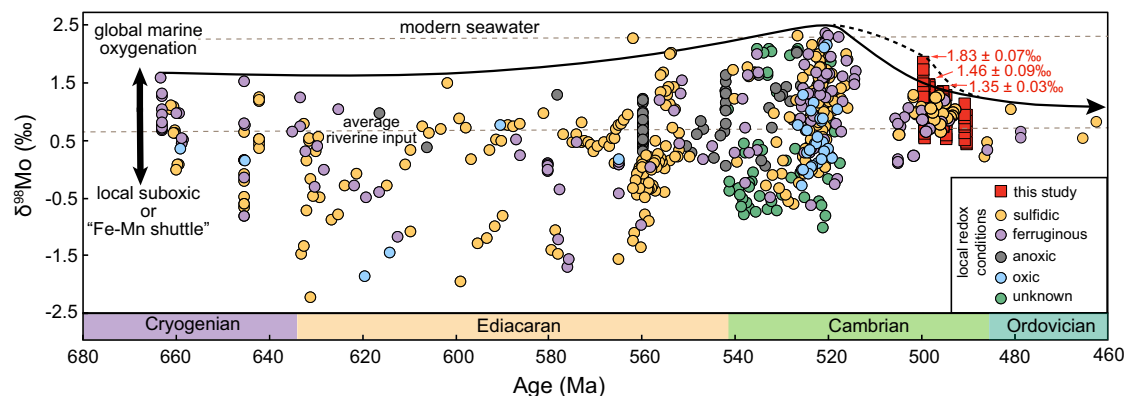


Fig. 4 | Late Neoproterozoic to early Paleozoic $\delta^{98}\text{Mo}$ records of global siliciclastic rocks. The black solid line represents a “best estimate” of coeval seawater $\delta^{98}\text{Mo}$ evolution, as constrained by $\delta^{98}\text{Mo}$ of marine, euxinic, and siliciclastic sedimentary rocks (data from compilation^{11,20} and this study).

pathway taking place inside the sediments at all times. Such a setting is also observed in parts of the modern Baltic Sea⁵⁸, where episodic inflows of oxygenated water are producing geochemical signatures similar to those in the Alum Shale.

The drivers for the MOE_{WC} events could be changes in ocean circulation, redox buffer supplies, and/or biological productivity. A recent modeling study showed that the continental configuration in the late Cambrian could have permitted oscillations at millennial scales between two globally distinct ocean circulation regimes with dramatic consequences for benthic O_2 levels worldwide (ranging between $>100\ \mu\text{mol kg}^{-1}$ and $<45\ \mu\text{mol kg}^{-1}$ in the abyssal oceans even if atmospheric pO_2 were at modern-day levels⁵⁹). Atmospheric pO_2 levels were likely lower in the Cambrian^{7,28} with the surface oceans containing likely only $\sim 10\text{--}40\%$ of today’s maximum O_2 concentration. Therefore, we envisage variations in the flux of oxygenated water into the Alum Shale basin, paced by shifts in ocean circulation, could have driven frequent switching between anoxic and oxic bottom water conditions in the region. Still, local sediment porewaters remained sulfidic at depth as evidenced by continuous pyrite formation at all times.

The Cambrian oceans were characterized by lower concentrations of redox buffers such as H_2S and Fe^{2+} , making millennial redox conditions sensitive to minor variations in the sulfide and iron contents at the basin scale. This inference is supported by a robust positive correlation ($R^2 = 0.84$, $n = 33$) between Mo concentrations and the degree of pyritisation (DOP_T) proxy $[(\text{S}/64)/(\text{Fe}/56)]$, with lower DOP_T values indicating non-euxinic conditions⁶⁰ (Supplementary Fig. 4). The redox landscape in this part of the late Cambrian oceans included anoxic conditions with either free H_2S (euxinic conditions) or Fe^{2+} (ferruginous conditions) prevailing in the water column, depending on the balance between H_2S production and Fe^{2+} supply²⁰. The MOE_{WC} events coincide with decreases of sedimentary Fe, S, and DOP_T values (Supplementary Fig. 3), indicating losses of both Fe^{2+} and H_2S from the water column. Therefore, we infer that water column redox conditions most likely transited from euxinic to oxic over millennial time-scales, and that these abrupt redox transitions were assisted or amplified by nonlinear positive feedbacks within the internal iron and sulfur cycles⁶⁰.

In addition, MOE_{WC} events could be linked to reduced biological productivity. The lower sedimentary content of the biolimiting nutrient P during the MOE_{WC} indicates decreased biological production. We suggest that low sedimentary P concentrations were caused by reduced P recycling within the basin, as this also explains the delayed P recovery after the basin returned to euxinic conditions (Supplementary Fig. 3). Specifically, the benthic dissolved P flux would cease in oxic sediments through adsorption onto FeOOH particulates present. The recurring anoxic settings promote the P desorption from FeOOH minerals⁶¹, allowing P diffusion upward across the anoxic/oxic boundary. If this cycling dominates the P supply in the local basin, then the sedimentary P recovering trend awaits the slow rise in local P inventory, as seen in the less sharp increase of

sedimentary P concentration relative to other redox proxies following MOE_{WC} events (Supplementary Fig. 3). In this view, MOE_{WC} periods appear to have influenced the nutrient availability in the basin even for a period after the millennial oxygenation events, and thus may have caused variation in photosynthetic productivity, organic matter remineralization in the water column, and organic matter export to sediments.

Millennial-scale release of hydrogen sulfide into surface waters (MDEs)

Given the long residence time of U and Mo in the Cambrian Ocean, faster and more complete removal during periods of intensified euxinia likely caused decreasing elemental concentrations in the sedimentary deposits, akin to that observed in restricted basins today. Widespread oceanic anoxia during the late Cambrian^{19,20,62} is characterized by lower U isotope values and reduced enrichments of both U and Mo, not only in the Scandinavian Alum Shale Basin^{10,20} but also in globally distributed records^{63,64}. Multiple lines of evidence suggest the Alum Shale was always well connected to the open ocean, for example, modestly high Mo/TOC ratios indicate that the basin was no more than moderately restricted (e.g., comparable to modern Cariaco basin, or perhaps the Framvaren Fjord)^{10,65}. Neodymium isotope evidence suggests the seawater mixing from the southeast Iapetus Ocean was a constant factor throughout Cambrian–Ordovician times⁶⁵. And finally, Mo, U, and sulfate must have been delivered from the open oceans to maintain euxinic conditions in large parts of the Alum Shale basin for most of the time. That said, periodic shoaling of the chemocline could have moved H_2S -bearing waters upwards in the water column, producing signatures similar to those observed in restricted basins, as shown in MDE scenarios C22E12 and C10E14 (Fig. 3FG). The strong negative covariation between $\delta^{98}\text{Mo}$ and $\delta^{238}\text{U}$, e.g., for the C22E12 ($r = -0.8$, $n = 12$) with a slope of -1.3 (Supplementary Fig. 5), is similar to the negative regression line (slope ~ -1.5) observed in modern restricted euxinic environments^{33,66}, suggesting MDEs represent periods with more complete Mo and U removal in the basin. That said, millennial-scale variations in water exchange (e.g., basinal restriction) likely contributed to variable H_2S levels in the basin that ultimately were modulated by ocean circulation regime through wind stress, relative sea level, and seafloor bathymetry.

The $\delta^{98}\text{Mo}$ and $\delta^{238}\text{U}$ co-variation during MDE events offers an opportunity to estimate the compositions of open ocean seawater. Applying the model proposed by Lu et al.³² yields the following estimates for coeval seawater: early Guzhangian (~ 500 Ma, C22E12) seawater would have had $\delta^{238}\text{U}$ and $\delta^{98}\text{Mo}$ values of -0.81 to -0.34‰ and $>1.46\text{‰}$, respectively. Late Paibian (~ 495 Ma, C10E14) seawater would have had $\delta^{238}\text{U}$ and $\delta^{98}\text{Mo}$ compositions of -0.83 to -0.67‰ and $>1.35\text{‰}$, respectively (Supplementary Fig. 5). The estimated seawater $\delta^{238}\text{U}$ values are consistent with published $\delta^{238}\text{U}$ data from upper Cambrian marine carbonates (-0.76 to -0.20‰)⁶³ assuming those carbonates recorded seawater $\delta^{238}\text{U}$ composition with a variable offset of $+0.27 \pm 0.28\text{‰}$ (2 SD)⁵². Also, the estimated seawater

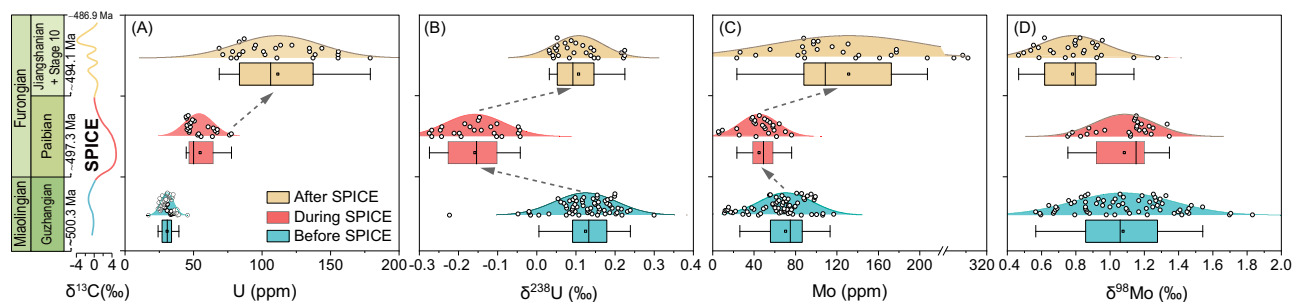


Fig. 5 | Statistical comparison ocean redox proxy data across the SPICE event. Distributions of (A) U concentrations, (B) $\delta^{238}\text{U}$ values, (C) Mo concentrations, and (D) $\delta^{98}\text{Mo}$ values for ten millennial redox events before, during, and after the SPICE event. The dashed black line within each panel represents the

median value of that distribution. Before the SPICE event: C16E7, C22E9, C22E12 and C23E10; during the SPICE event: C10E14 and C12E12; after the SPICE event: C1E5, C1E6, C9E6 and C3E7. See stratigraphic levels of these ten events in Fig. 1C.

$\delta^{98}\text{Mo}$ is comparable to the maximum $\delta^{98}\text{Mo}$ value of $1.33 \pm 0.12\%$ reported from coeval euxinic shales²⁰ (Fig. 4).

The implication is that anoxic (and euxinic) conditions prevailed in late Cambrian oceans, as reflected by significantly lower oceanic $\delta^{238}\text{U}$ and $\delta^{98}\text{Mo}$ values than in the modern oceans (-0.39% and 2.34% , respectively). We observe a single high peak $\delta^{98}\text{Mo}$ value, reaching up to 1.83% (C23E10 despite a noisy trend; Supplementary Fig. 2E) in the early Guzhangian. This is higher than the estimate of $>1.46\%$ for contemporaneous seawater and may suggest a higher coeval seawater value (Fig. 4). Alternatively, the elevated sedimentary $\delta^{98}\text{Mo}$ values may represent a temporary seawater $\delta^{98}\text{Mo}$ overshoot caused by a “reservoir effect”, wherein preferential removal of light Mo isotopes outpaces the basin’s Mo supply⁶⁷. This effect occurs when Mo burial with isotope fractionation is amplified in the euxinic basin, preferentially removing lighter Mo isotopes and leaving behind heavier isotopic residue in the basin for a short period. Consequently, the average $\delta^{98}\text{Mo}$ value in the local basin can temporarily exceed that of the coeval open ocean. Therefore, we urge caution when using Mo-U isotope trends during plausible MDEs like C23E10 to gauge the composition of open ocean seawater. In any case, the consistently lower $\delta^{98}\text{Mo}$ values observed during all MDEs, compared to modern seawater and restricted black shales, provide compelling evidence for a more extensively euxinic global ocean state during the late Cambrian than exists today (Fig. 4).

Relationship between local MREs and Myr-scale global redox and biotic changes

A major carbon cycle perturbation event in the late Cambrian, termed the Steptoean Positive Carbon Isotope Excursion (SPICE, from ~ 497.5 to ~ 494.5 Ma), covers a period of extreme oxygen deficiency in the global oceans^{10,19,63}. In line with this global trend, the $\delta^{238}\text{U}$ distributions of all ten MREs before, during, and after the SPICE event show a clear negative shift (Fig. 5), reflecting enhanced U burial in anoxic sediments worldwide^{20,36,63}. These observations suggest that, although the studied MREs reflect local redox fluctuations, they also bear evidence of a changing global ocean redox state. In contrast, the $\delta^{98}\text{Mo}$ records show no obvious trend across the SPICE event, implying that the $\delta^{98}\text{Mo}$ fractionations derived from H_2S fluctuations within the Alum Shale Sea had a greater influence on the sedimentary $\delta^{98}\text{Mo}$ than changes in the open ocean composition. The MOE_{WC} events occurred more frequently after the SPICE event (Fig. 3), aligning with the trend towards more oxygenated oceans globally in the aftermath of the SPICE event^{10,62}. This comparison demonstrates the potential of submillimeter-scale geochemical analyses in uncovering connections between short-term local and long-term global redox variations in ancient oceans.

The millennial redox trends align with changes in animal diversification in the Furongian of the Baltic Alum Shale as well as in global compilations (Fig. 3B, C). Although only ten events were studied here, the intermittent MOE_{WC} and MDE events during the Miaolingian shifted to predominantly MDE events during the early Furongian SPICE event. These changes coincided with a significant decline in diversity in Baltica,

reaching its lowest point. After the SPICE event, MOE_{WC} events gradually took over, each linked to a recovery in biodiversity in Baltica (Fig. 3C). Also, the elevated occurrences of local MOE_{WC} through the Furongian correlate well with increases in global biotic richness and relative diversification rates (Fig. 3B). We suggest that the MOE_{WC} events potentially created seafloor niches for aerobic, opportunistic animals. These events extended for ~ 650 – 2250 yrs, which must have been long enough for pioneer communities to colonize the seafloor and for faunal recovery, because it typically takes less than two years for mature metazoan and macrofaunal communities, and only days or weeks for small, opportunistic meiofaunal animals to colonize the seafloor⁶⁸. Both trilobites and brachiopods (such as *Orusia lenticularis*) living in the Alum Shale Sea had planktonic larvae stage⁶⁹, enabling them to migrate and survive the harsh living conditions in the Cambrian oceans. In addition, the oxygen demands of Cambrian organisms were likely lower⁷⁰. Therefore, the MOE_{WC} events have allowed opportunistic benthic communities to flourish intermittently before becoming locally or regionally extirpated when anoxic conditions re-emerged.

Materials and methods

Samples

Billegrav-2 is a fully cored scientific well with a total depth of 125.9 m, made by the Geological Survey of Denmark and Greenland (GEUS) and the University of Copenhagen on southern Bornholm, Denmark. The Alum Shale Formation was encountered between 95.4 and 122.2 m. For the geological background of the Alum Shale, see Supplementary Note 1. The Billegrav-2 core was split through the core center (55 mm core diameter) at ten shale intervals, including MREs (Fig. 1C). The ten 4–10 cm-long core pieces all demonstrate significant Mo concentration declines and/or contain benthic animal fossils.

XRF core scanning

To accurately locate the events, ten core pieces were scanned for elemental profiles at a stratigraphic resolution of 0.2 mm using an Itrax X-ray fluorescence core scanner (XRF-CS) at the Globe Institute, University of Copenhagen¹⁵. The measurements were performed on the split interior, plain, cleaned core surface. The applied voltage and current of Rh energy are 30 kV and 50 mA, respectively, and the exposure time is 100 s. Absorption peaks were identified and peak areas were quantified using the Q-Spec software CoreScanner 8.6.4 Rh from Cox Analytical Systems. Elemental concentrations were first exported with Peak Area - Concentration relationship obtained with factory settings using the SGR-1 standard (Green River Shale). To correct $\text{Mo}_{\text{XRF-CS}}$, we applied documented calibration for the Alum Shale ($\text{Mo}_{\text{ID-ICP-MS}} = 2 \times \text{Mo}_{\text{XRF-CS}} - 5 \text{ ppm}^{15}$; Supplementary Fig. 6A). The deviation of the slope from unity highlights the need for independent control measurements to ensure accurate trace element concentration profiles from XRF-CS. All XRF-CS Mo data reported here have been corrected using this calibration. A good correlation was also found

between U concentrations obtained in XRF-CS and the isotope dilution data, $U_{ID-ICP-MS} = 2.6 \times U_{XRF-CS} + 45$ ppm (Supplementary Fig. 6B).

Submillimeter-scale U-Mo isotopes

To explore the MREs further using metal isotope analysis, we cut 127 ultra-narrow slots (spaced as closely as 0.25 mm apart) from the ten identically sized shale pieces along their bedding plane using a 0.25 mm-thick diamond wire, and collected the resulting powder from each slot for analyses. The shale powder was completely digested in concentrated hydrochloric and nitric acid following the procedure described by Dickson et al.³⁴. Uranium and molybdenum isotopes and their concentrations were analyzed by MC-ICP-MS at Royal Holloway University of London, UK. The sample solutions were spiked using ^{97}Mo – ^{100}Mo and IRMM 3636a ^{233}U – ^{236}U and purified using AG1-X8 anion resin and UTEVA resin. Uncertainties for individual $\delta^{98}\text{Mo}$ analyses are propagated from the sample standard error and the bracketing standards (NIST 3134). Uncertainties for $\delta^{238}\text{U}$ are the two-fold standard deviations of long-term external reproducibility of USGS Devonian Ohio Shale (SDO-1, $\pm 0.07\%$, 2 SD, $n = 9$) over the three months of Alum Shale measurements. For more details, see Supplementary Note 2. The results are listed in Supplementary Table 1 and Supplementary information.

Animal diversity

Species-level richness in each biozone from the Baltic Alum Shale Formation is based on refs. 21,25,71 with a few adjustments according to refs. 72,73. The ages for the lower boundary of biozones are based on ref. 36. The Baltic richness data are listed in Supplementary Table 2. The global generic richness data are from refs. 74,75.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data are available in the article and/or the Supplementary Information. The original geochemical data are publicly available at Zenodo (<https://doi.org/10.5281/zenodo.16810480>).

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Author contributions

T.W.D. conceived the idea. T.W.D. and Z.Z. designed the study. Z.Z., A.T.N., N.H.S., T.W.D., and I.W.H. collected the samples. Z.Z., I.W.H., A.J.D., and A.B. performed the elemental and isotopic analyses. A.T.N. compiled the species-level richness dataset. T.W.D. and C.Z. supervised the research and guided data interpretation. Z.Z., Z.J., P.J., and Z.G. prepared the data visualizations. S.D., C.Z., T.W.D., Z.J., and Z.Z. acquired funding for the project. Z.Z. wrote the initial manuscript draft with contributions from T.W.D., A.J.D., C.Z., S.D., A.T.N., Z.J., P.J., Z.G., A.B., N.H.S., and I.W.H. All authors contributed substantially to data acquisition, interpretation, and discussion of the manuscript content.

Competing interests

The authors declare no competing interests.

Additional information

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